AIML ZG546 - Software Engineering for Machine Learning

Assignment II

BrakeGuard AI - Implementation, Code Quality and Testing of an ML-enabled Predictive-Maintenance System

Group No: 14

Sl. No	BITS ID	Name	Contribution of team member (Qualitative)	Percentage Contribution (out of 100)
1	2025ae05802	SAI KRISHNA VIJAYA RAM K	Refactoring into the brakeguard package (data, features, models, pipeline); research-vs-production comparison (Tasks 1, 2)	100%
2	2025ae05999	S. S SRINIVAS RAO	FastAPI REST API: schemas, status codes, error envelope, API-key security, dashboard (Task 5)	100%
3	2025ae05192	ABHISHEK ARORA	Test strategy and pytest suites: unit, data-quality, ML training/inference, integration; coverage (Tasks 6, 7)	100%
4	2025ae05200	RAJESH. R. SHENOI	Logging and error handling, black/isort/flake8/pylint, quality metrics and drift, production testing and security, report (Tasks 3, 4, 8, 9)	100%

Code: `BrakeGuard_AI_Assignment_Application/brakeguard_ai-3` (builds on the Assignment I prototype in `brakeguard_ai-2`). Implementation notebook: `notebooks/Group14.ipynb`.

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0. System overview

BrakeGuard AI predicts the probability that a vehicle's brake system will develop a fault within the next 30 days / 1,000 km, using six telemetry signals: rotor temperature, line pressure, pad wear, caliper vibration, brake-fluid age and odometer. A Random-Forest classifier produces a probability, which is mapped to a LOW / MEDIUM / HIGH risk band (thresholds 0.40 / 0.70) with an explanation of the main drivers. The ML component sits inside non-ML components: a REST API, a feature store, an event bus for telemetry, an audit log and drift monitoring.

Assignment I delivered a working prototype (`brakeguard_ai-2`) with module-level globals, no data validation, one happy-path test file, no logging, and a dashboard bug (the audit table read `fault_probability` while the log stored `probability`). Assignment II rebuilds it as the production-style package below. The model family and hyper-parameters are kept, so the comparison isolates software-engineering changes.

Package / module	Responsibility	Key classes / functions
`brakeguard/config.py`	Immutable, environment-overridable settings	`Settings`, `RiskThresholds`, `ModelHyperParams`
`brakeguard/schemas.py`	Single source of truth for the telemetry contract	`TELEMETRY_SCHEMA`, `FeatureSpec`
`brakeguard/data/`	Data ingestion; data-quality validation	`TelemetryDataSource` (ABC), `SyntheticTelemetrySource`, `CSVTelemetrySource`, `DataValidator`, `DataQualityReport`
`brakeguard/features/`	Feature engineering; online feature store	`FeatureEngineer` (sklearn transformer), `FeatureStore`
`brakeguard/models/`	Training; versioned persistence; inference	`ModelTrainer`, `ModelRegistry`, `ModelBundle`, `BrakeFaultPredictor`, `Prediction`
`brakeguard/evaluation/`	Model-quality metrics	`evaluate_classifier`, `expected_calibration_error`
`brakeguard/monitoring/`	Audit log, pub/sub, drift	`AuditLog`, `EventBus`, `DriftDetector`, `population_stability_index`
`brakeguard/pipeline.py`	Orchestration with quality gates	`TrainingPipeline`, `load_or_train`
`brakeguard/api/`	REST layer	`create_app`, routes, schemas, error handlers, API-key security, `ServiceContainer`
`scripts/`	CLIs	`train.py`, `quality_report.py`, `build_notebooks.py`

Objective 1 - Implementation and code sharing

Task 1 - OOP / functional design of the ML application

The code is organised around the ML lifecycle, with one class per responsibility. The design follows these principles:

- **Single responsibility:** ingestion, validation, feature engineering, training, persistence and inference are separate classes. Each can be tested on its own (Section 6) and replaced independently.
- **Dependency inversion:** `TrainingPipeline` depends on the abstract `TelemetryDataSource`, so synthetic data, a CSV export or a database are interchangeable. The integration test trains from a CSV without code changes.
- **Composition over globals:** the API builds a `ServiceContainer` (predictor, feature store, event bus, audit log, drift detector) per application instance through the `create_app` factory. Tests create isolated apps with no model or with an API key, without shared mutable state.
- **Functional core:** metrics (`evaluate_classifier`, `expected_calibration_error`, `population_stability_index`) are pure functions of their inputs, and settings and value objects are frozen dataclasses.
- **Training/serving consistency:** `FeatureEngineer` is an sklearn transformer placed inside the model `Pipeline`. The same fitted transformation, including imputation medians and the derived features `pressure_deviation`, `thermal_wear_index` and `fluid_age_years`, is serialised with the model and replayed at inference time.

The training workflow with its two quality gates (data quality before training, model quality before registration):

brakeguard/pipeline.py - TrainingPipeline.run

```
    def run(self, register: bool = True) -> ModelBundle:
        """Execute the pipeline and (optionally) register the model."""
        logger.info("Training pipeline started (model version %s)", self.settings.model_version)
        frame = self.source.load()
        quality = self.validator.validate_or_raise(frame, require_target=True)
        result = self.trainer.train(frame)
        self._check_quality_gates(result.test_metrics.to_dict())
        bundle = ModelBundle(
            pipeline=result.pipeline,
            version=self.settings.model_version,
            metrics=result.test_metrics.to_dict(),
            feature_names=list(ENGINEERED_FEATURES),
            reference_frame=result.reference_frame.sample(
                n=min(1000, len(result.reference_frame)), random_state=self.settings.random_seed
            ).reset_index(drop=True),
            data_quality=quality.to_dict(),
        )
        if register:
            self.registry.save(bundle)
        logger.info("Training pipeline finished")
        return bundle
```

brakeguard/features/engineering.py - FeatureEngineer.transform

```
    def transform(self, X: pd.DataFrame) -> pd.DataFrame:  # noqa: N803
        """Return a frame with raw + derived features, imputed."""
        if not hasattr(self, "medians_"):
```

```
            raise FeatureEngineeringError("FeatureEngineer must be fitted before transform")
        frame = self._check_input(X).copy()
        n_missing = int(frame.isna().sum().sum())
        if n_missing:
            logger.warning("Imputing %d missing value(s) with training medians", n_missing)
            frame = frame.fillna(value=self.medians_)
        frame["pressure_deviation"] = (frame["brake_pressure"] - self.nominal_pressure).abs()
        frame["thermal_wear_index"] = frame["brake_temperature"] * frame["pad_wear_pct"] / 100.0
        frame["fluid_age_years"] = frame["fluid_age_days"] / 365.0
        return frame[list(ENGINEERED_FEATURES)]
```

Task 2 - Research code vs production code

Component compared: **model training and inference**. The research version is `notebooks/01_research_prototype.ipynb`, written as typical exploratory cells with the same logic as Assignment I's `model_service.py`. The production version is the `brakeguard` package plus the `scripts/train.py` CLI. The notebook is kept on purpose, and its outputs show two defects the production code removes:

- Its explanation heuristic (`importance x raw value`) always ranks `odometer_km` first (score 7,681 vs 39 for the next feature) because odometer values are in tens of thousands of km. Production uses `importance x |z-score|`, and the top drivers for BG-10245 become thermal_wear_index, pad_wear_pct and vibration_rms.
- It scores physically impossible input (`pad_wear_pct = 450` gives p = 0.94). Production rejects this with HTTP 422.

Research (notebook cell)

```
# research notebook - globals, magic numbers, implicit feature order, prints
model = RandomForestClassifier(n_estimators=160, max_depth=8, min_samples_leaf=4,
random_state=42, class_weight="balanced")
model.fit(X_train, y_train)
print("recall", recall_score(y_test, pred))
joblib.dump(model, "model_research.joblib")          # relative path, no metadata
vehicle = [186.4, 72.1, 78.0, 3.2, 410, 84250]       # which number is which?
prob = model.predict_proba([vehicle])[0, 1]
risk = "HIGH" if prob >= 0.70 else "MEDIUM" if prob >= 0.40 else "LOW"
```

Production (package API)

```
# production - typed, validated, logged, versioned, tested
settings = get_settings()                                  # env-overridable, frozen
bundle = TrainingPipeline(settings=settings).run()         # validate -> train -> gate ->
register
predictor = BrakeFaultPredictor(bundle, settings.thresholds)
prediction = predictor.predict(feature_store.get("BG-10245"))   # dict keyed by schema names
prediction.risk, prediction.fault_probability, prediction.top_factors
```

Concern	Research notebook	Production package
Structure	Linear cells and global variables; results depend on execution order	Classes with single responsibilities, wired by `TrainingPipeline`
Configuration	Magic numbers and hard-coded	`Settings` dataclass + `BRAKEGUARD_*`

Concern	Research notebook	Production package
	paths	environment variables
Data contract	Implicit NumPy column order	`TELEMETRY_SCHEMA` shared by validator, model and API
Data quality	None	`DataValidator` gate (schema, missing, range, labels, duplicates)
Model acceptance	Human reads printed metrics	Automatic gate: recall >= 0.75, PR-AUC >= 0.70
Persistence	Bare model, relative path	`ModelBundle`: version, metrics, timestamp, sklearn version, drift reference
Errors / logs	Tracebacks / print	Typed exceptions -> HTTP codes; `logging` with levels
Tests / style	None	96 pytest tests (97 % coverage); pylint 10.00/10, flake8 0

Task 3 - Error handling and logging

All modules log through the `brakeguard.*` logger hierarchy configured in `logging_config.py` (format `time | level | module | message`). Levels have a fixed meaning: **INFO** for normal lifecycle events, **WARNING** for degraded but recoverable situations, and **ERROR** for failed operations where the caller receives an error. Failures raise typed exceptions from `exceptions.py`, which `api/errors.py` maps to HTTP status codes, so the API never returns a raw stack trace.

Critical function	INFO	WARNING	ERROR / exception
`CSVTelemetrySource.load`	rows loaded	file has no rows	file missing / unparsable -> `DataIngestionError`
`DataValidator.validate_or_raise`	validation passed	duplicates; missing values below gate; issues found	data set rejected -> `DataValidationError`
`FeatureEngineer.transform`	-	missing values imputed with training medians	missing / non-numeric columns -> `FeatureEngineeringError` (422)
`ModelTrainer.train`	rows, positive rate, hold-out metrics	-	single-class target / fit failure -> `ModelTrainingError`
`TrainingPipeline` quality gate	model passed gates	-	model rejected -> `ModelTrainingError`, nothing registered
`ModelRegistry.load` / `load_or_train`	model loaded / registered	sklearn version mismatch; artefact missing -> retrain fallback	corrupt artefact -> `ModelRegistryError`
`BrakeFaultPredictor.predict*`	records scored	-	no model -> `ModelNotReadyError` (503)
`FeatureStore.get`	vehicle inserted /	unknown vehicle	`VehicleNotFoundError`

Critical function	INFO	WARNING	ERROR / exception
	updated		(404)
`EventBus.publish`	subscriber registered	event with no successful subscriber	failing subscriber logged with traceback; others still run
`AuditLog.record` / `DriftDetector`	drift summary	HIGH-risk prediction; PSI >= 0.10 / 0.25	-
HTTP middleware / handlers	method, path, status, latency per request	422 / 404 responses	unexpected exception -> generic 500 with request id

brakeguard/models/registry.py - error handling with meaningful levels

```
    def load(self) -> ModelBundle:
        """Load a bundle; raises :class:`ModelRegistryError` on any problem."""
        if not self.exists():
            raise ModelRegistryError(f"No model artefact at {self.path}")
        try:
            bundle = joblib.load(self.path)
        except Exception as exc:  # joblib can raise many unpickling errors
            logger.error("Model artefact %s is corrupt: %s", self.path, exc)
            raise ModelRegistryError(f"Corrupt model artefact: {exc}") from exc
        if not isinstance(bundle, ModelBundle):
            logger.error("Artefact %s has unexpected type %s", self.path, type(bundle).__name__)
            raise ModelRegistryError("Artefact is not a ModelBundle")
        if bundle.sklearn_version != sklearn.__version__:
            logger.warning(
                "Model trained with scikit-learn %s, running %s - consider retraining",
                bundle.sklearn_version,
                sklearn.__version__,
            )
        logger.info("Loaded model %s (trained %s)", bundle.version, bundle.trained_at)
        return bundle
```

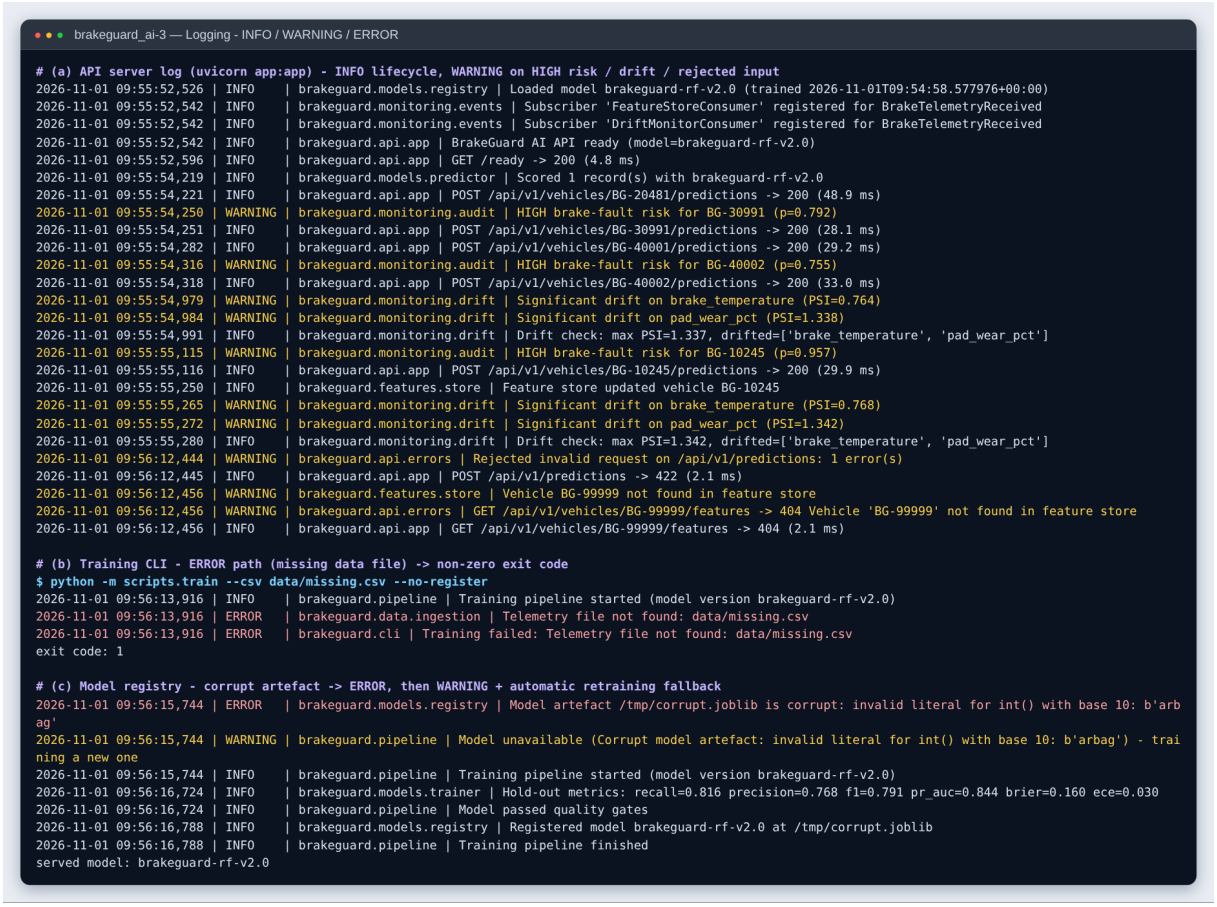


Figure 1 - Real log output: INFO lifecycle, WARNING for HIGH risk / drift / rejected input, ERROR for a missing data file (CLI exits with code 1) and a corrupt model artefact (automatic retraining fallback).

Task 4 - Code formatting and linting

Tools: **black** (formatting, line length 110), **isort** (import order, black profile), **flake8** (PEP 8 / pyflakes) and **pylint** (static analysis). Configuration is in `pyproject.toml` and `.flake8`. The same commands run as a build gate in the Dockerfile. Two "before" baselines were captured: the Assignment I code base, and the new package before black and isort were run and findings fixed.

Check	Assignment I code (before)	New package before fixes	Final package (after)
black	6 of 8 files would be reformatted	7 of 44 files would be reformatted	all 46 files unchanged
isort	3 files incorrectly sorted	clean	clean
flake8	37 issues (E231 x15, E302 x13, E501 x7, E305 x2)	12 issues (E501)	0 issues
pylint	7.25 / 10 (missing docstrings, long lines, too many args, ...)	9.83 / 10	10.00 / 10

Fixes applied to the package after the baseline: automatic reformatting (black) and import sorting (isort); renaming `VehicleId` to `VEHICLE_ID_PATH` (constant naming); removing variable shadowing (`audit` -> `entry`); marking intentionally unused arguments; and documenting why sklearn's `fit()`-time attributes and

the dataclass field counts are acceptable. Suppressions are local and commented, never global. Raw reports are in `reports/lint/`.

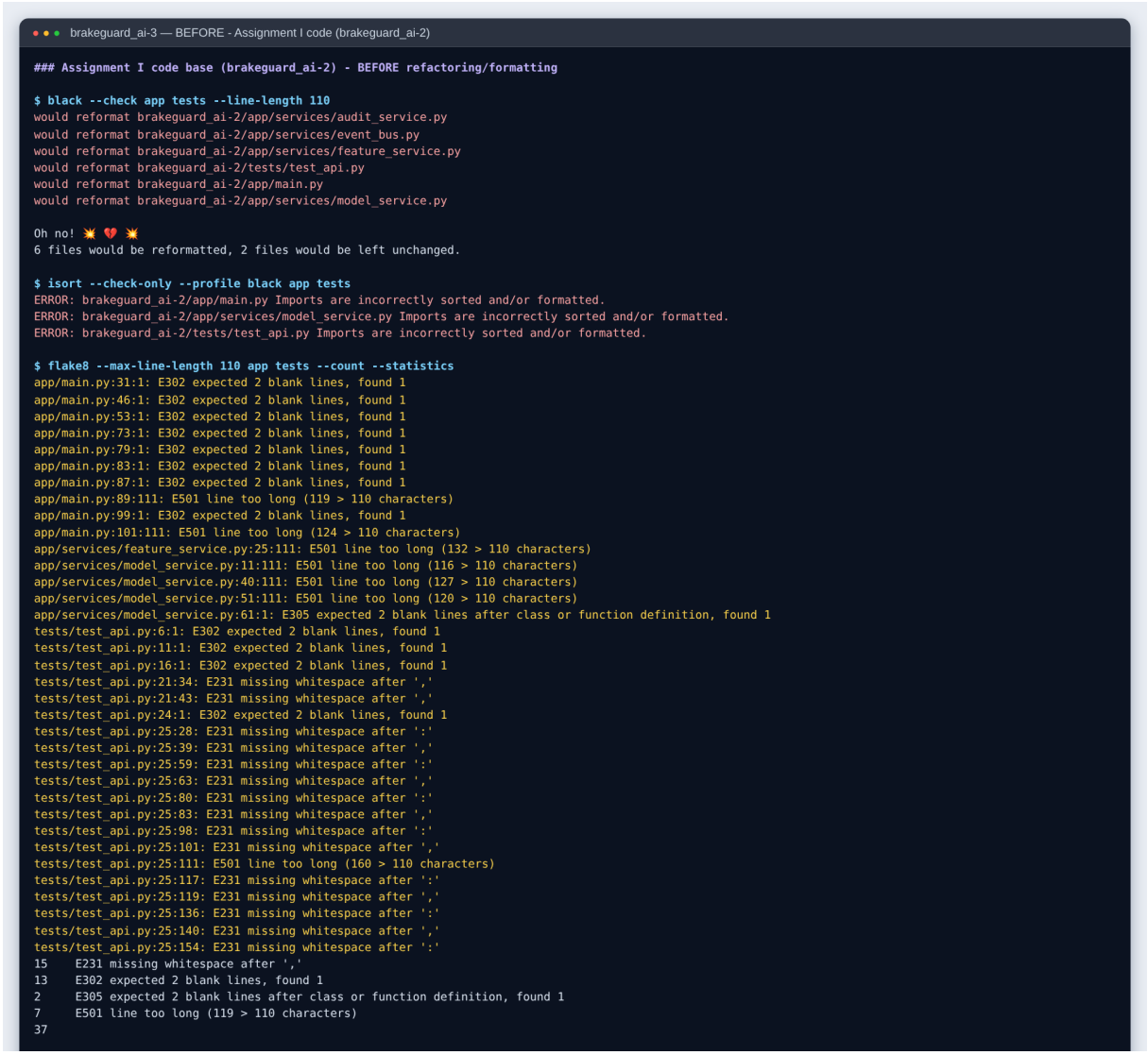
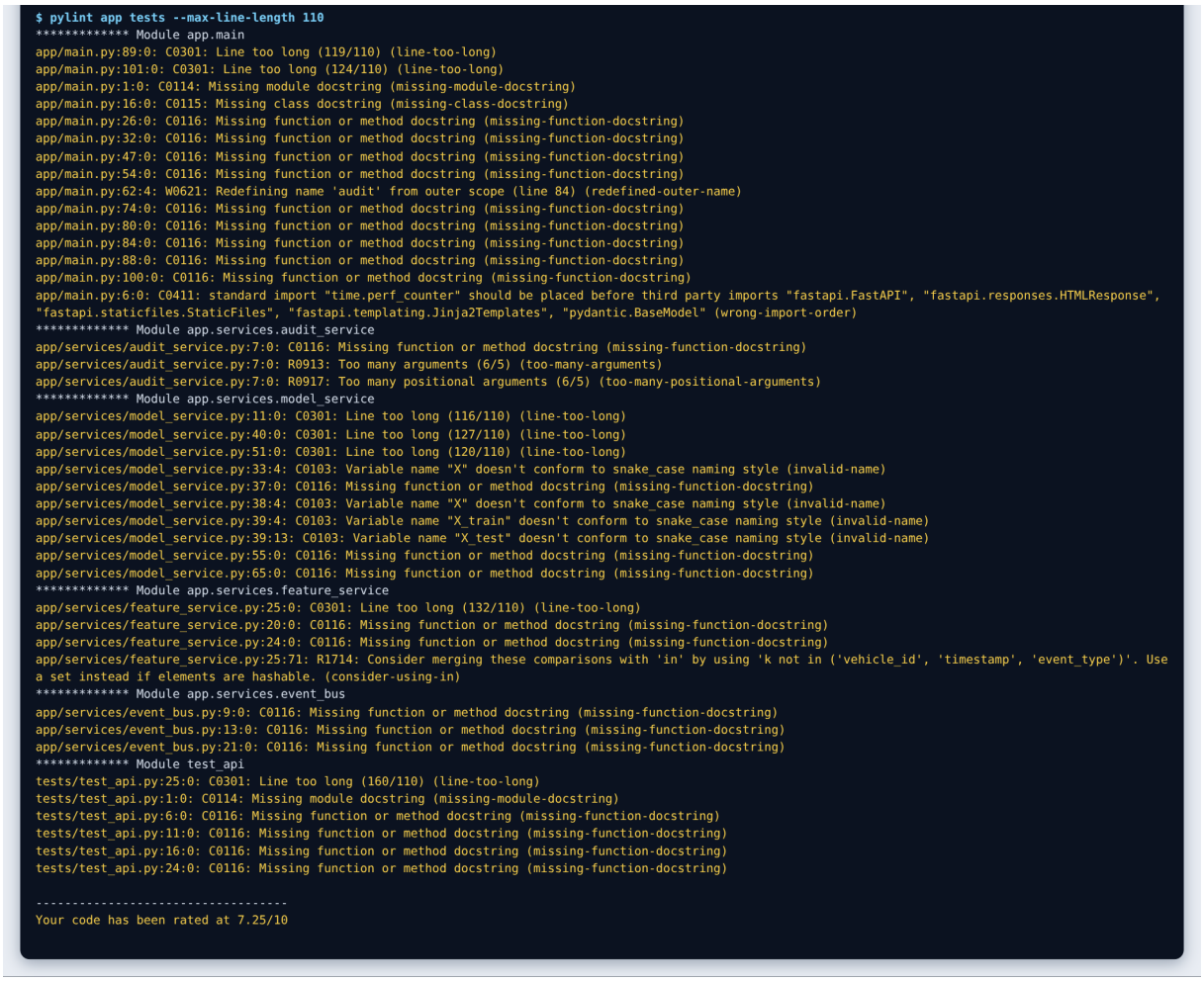


Figure 2a - BEFORE: black, isort and flake8 on the Assignment I code base.



```
$ pylint app tests --max-line-length 110
************* Module app.main
app/main.py:89:0: C0301: Line too long (119/110) (line-too-long)
app/main.py:101:0: C0301: Line too long (124/110) (line-too-long)
app/main.py:1:0: C0114: Missing module docstring (missing-module-docstring)
app/main.py:16:0: C0115: Missing class docstring (missing-class-docstring)
app/main.py:26:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:32:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:47:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:62:4: W0621: Redefining name 'audit' from outer scope (line 84) (redefined-outer-name)
app/main.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:80:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:84:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:88:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:100:0: C0116: Missing function or method docstring (missing-function-docstring)
app/main.py:6:0: C0411: standard import "time.perf_counter" should be placed before third party imports "fastapi.FastAPI", "fastapi.responses.HTMLResponse",
"fastapi.staticfiles.StaticFiles", "fastapi.templating.Jinja2Templates", "pydantic.BaseModel" (wrong-import-order)
************* Module app.services.audit_service
app/services/audit_service.py:7:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/audit_service.py:7:0: R0913: Too many arguments (6/5) (too-many-arguments)
app/services/audit_service.py:7:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
************* Module app.services.model_service
app/services/model_service.py:11:0: C0301: Line too long (116/110) (line-too-long)
app/services/model_service.py:40:0: C0301: Line too long (127/110) (line-too-long)
app/services/model_service.py:51:0: C0301: Line too long (120/110) (line-too-long)
app/services/model_service.py:33:4: C0103: Variable name "X" doesn't conform to snake_case naming style (invalid-name)
app/services/model_service.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/model_service.py:38:4: C0103: Variable name "X" doesn't conform to snake_case naming style (invalid-name)
app/services/model_service.py:39:4: C0103: Variable name "X_train" doesn't conform to snake_case naming style (invalid-name)
app/services/model_service.py:39:13: C0103: Variable name "X_test" doesn't conform to snake_case naming style (invalid-name)
app/services/model_service.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/model_service.py:65:0: C0116: Missing function or method docstring (missing-function-docstring)
************* Module app.services.feature_service
app/services/feature_service.py:25:0: C0301: Line too long (132/110) (line-too-long)
app/services/feature_service.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/feature_service.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/feature_service.py:25:71: R1714: Consider merging these comparisons with 'in' by using 'k not in ('vehicle_id', 'timestamp', 'event_type')'. Use
a set instead if elements are hashable. (consider-using-in)
************* Module app.services.event_bus
app/services/event_bus.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/event_bus.py:13:0: C0116: Missing function or method docstring (missing-function-docstring)
app/services/event_bus.py:21:0: C0116: Missing function or method docstring (missing-function-docstring)
************* Module test_api
tests/test_api.py:25:0: C0301: Line too long (160/110) (line-too-long)
tests/test_api.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests/test_api.py:6:0: C0116: Missing function or method docstring (missing-function-docstring)
tests/test_api.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
tests/test_api.py:16:0: C0116: Missing function or method docstring (missing-function-docstring)
tests/test_api.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)

---------------------------------
Your code has been rated at 7.25/10
```

Figure 2b - BEFORE: pylint on the Assignment I code base (7.25/10).



Figure 3 - BEFORE: the new package prior to black/isort and lint fixes (9.83/10, 12 flake8 issues).

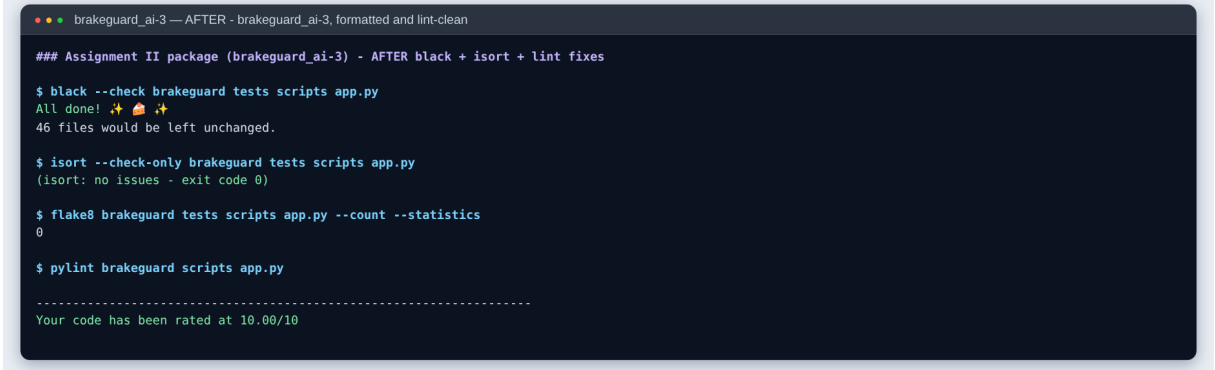


Figure 4 - AFTER: black and isort clean, flake8 0 issues, pylint 10.00/10.

Task 5 - REST API (FastAPI)

The API is created by `brakeguard.api.create_app()` and served by `uvicorn app:app`. It follows these design practices:

- **Resource-oriented, versioned URLs** under `/api/v1`, with nouns and HTTP verbs (`POST /predictions`, `GET /vehicles/{id}/features`). Operational probes (`/health` liveness, `/ready` readiness) are kept outside the versioned prefix.
- **Explicit request/response schemas** (Pydantic) published in OpenAPI/Swagger. Field ranges are derived from `TELEMETRY_SCHEMA`, unknown fields are forbidden, and batch size is limited to 1-500.
- **Meaningful status codes:** 200 OK, 202 Accepted (telemetry is processed asynchronously in production), 401 Unauthorized, 404 Not Found, 422 Unprocessable Content, 503 Service Unavailable (no model). All are documented per endpoint.
- **Uniform error envelope** `{"error": {"code", "message", "request_id", "details"}}` for every non-2xx response.
- **Traceability:** every response carries `X-Request-ID` and `X-Process-Time-ms`, and every prediction gets a `prediction_id` stored in the audit log with the model version.
- **Security:** optional `X-API-Key` authentication on `/api/v1/*` (Section 9).

Method	Endpoint	Purpose	Codes
GET	/health, /ready	Liveness / readiness probes	200 / 200, 503
GET	/api/v1/model	Model version, training time, metrics, thresholds	200, 401, 503
POST	/api/v1/predictions	Score telemetry in the request body	200, 401, 422, 503
POST	/api/v1/predictions/batch	Score up to 500 records (vectorised)	200, 401, 422, 503
GET	/api/v1/vehicles	List vehicles in the feature store	200, 401
GET	/api/v1/vehicles/{id}/features	Latest features of a vehicle	200, 401, 404, 422
POST	/api/v1/vehicles/{id}/predictions	Score a vehicle from the feature store	200, 401, 404, 503
POST	/api/v1/telemetry	Publish a BrakeTelemetryReceived event	202, 401, 422
GET	/api/v1/events, /api/v1/audit	Recent events / audited predictions	200, 401
GET	/api/v1/monitoring	Latency p50/p95, model quality, data drift	200, 401

brakeguard/api/routes.py - prediction endpoint

```
@api_router.post("/predictions", response_model=PredictionResponse, responses=ERRORS,
tags=["predictions"])
def predict(body: PredictionRequest, c: ServiceContainer = Depends(get_container)) ->
PredictionResponse:
    """Score telemetry supplied in the request body."""
    start = perf_counter()
    pred = c.predictor.predict(body.telemetry.model_dump())
    return _to_response(c, body.vehicle_id, pred, (perf_counter() - start) * 1000)
```

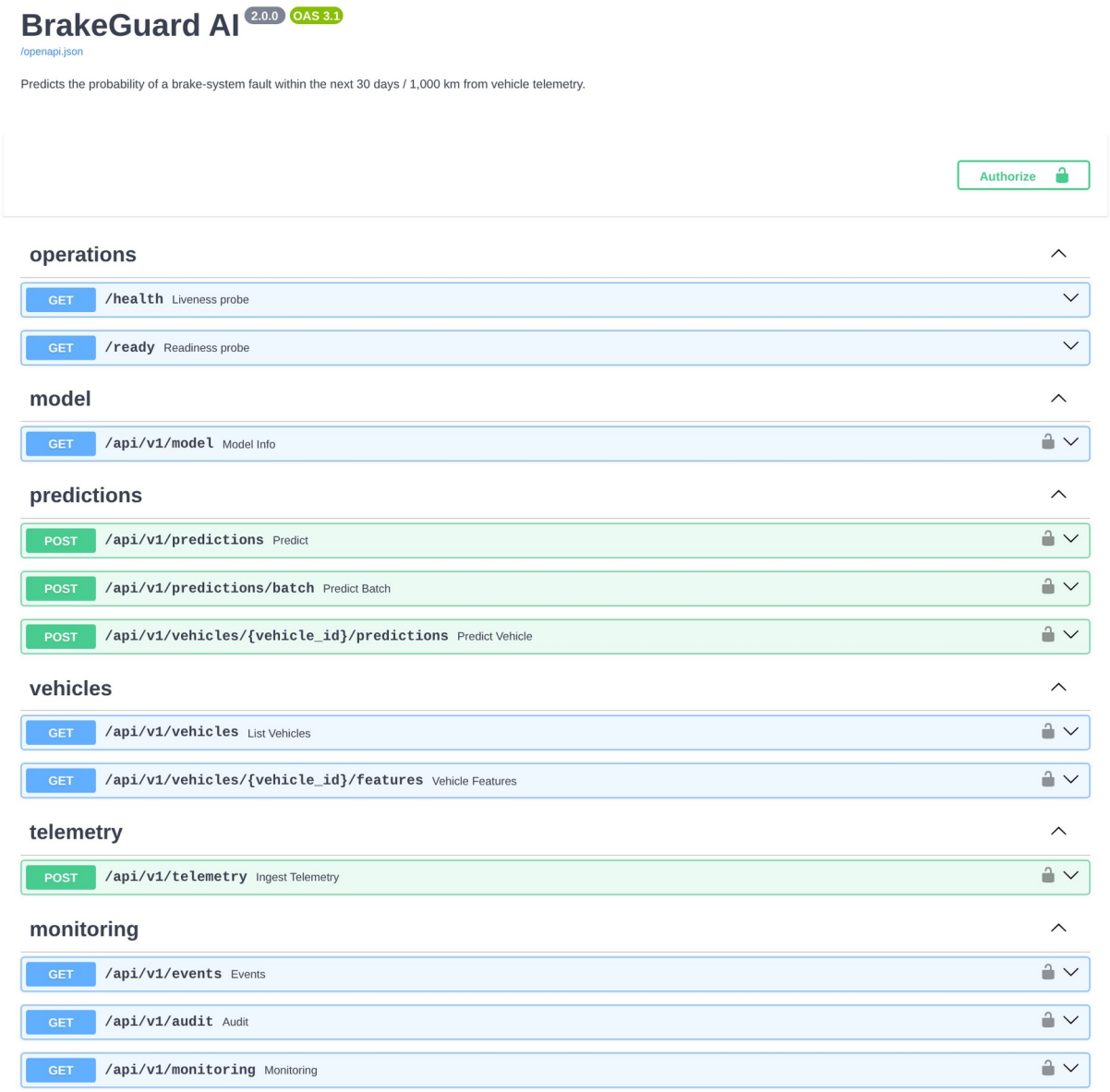


Figure 5 - Swagger UI (/docs): endpoints grouped by tag; lock icons mark API-key-protected routes.

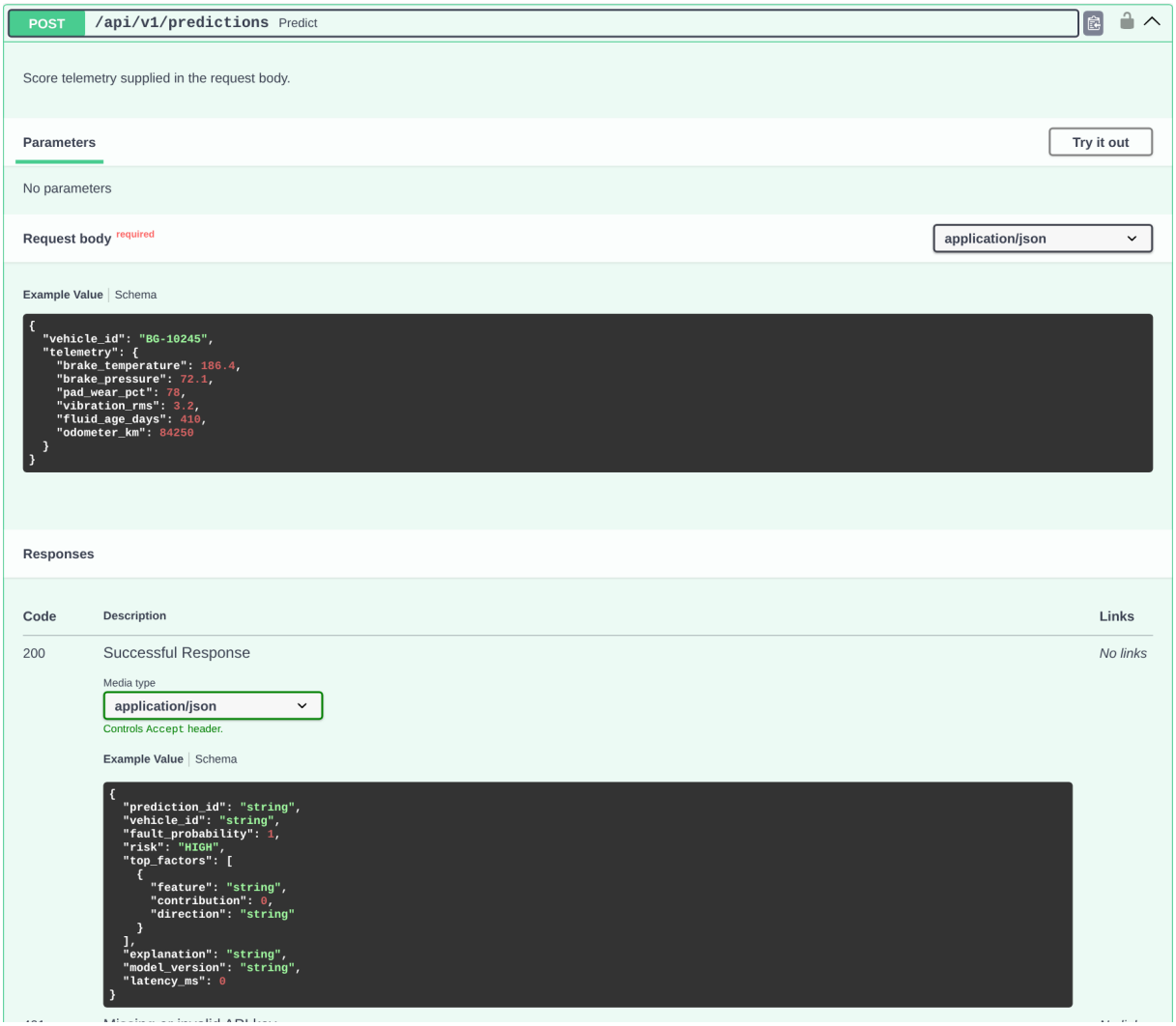


Figure 6 - Request schema with example, and documented 200/401/422/503 responses for POST /api/v1/predictions.

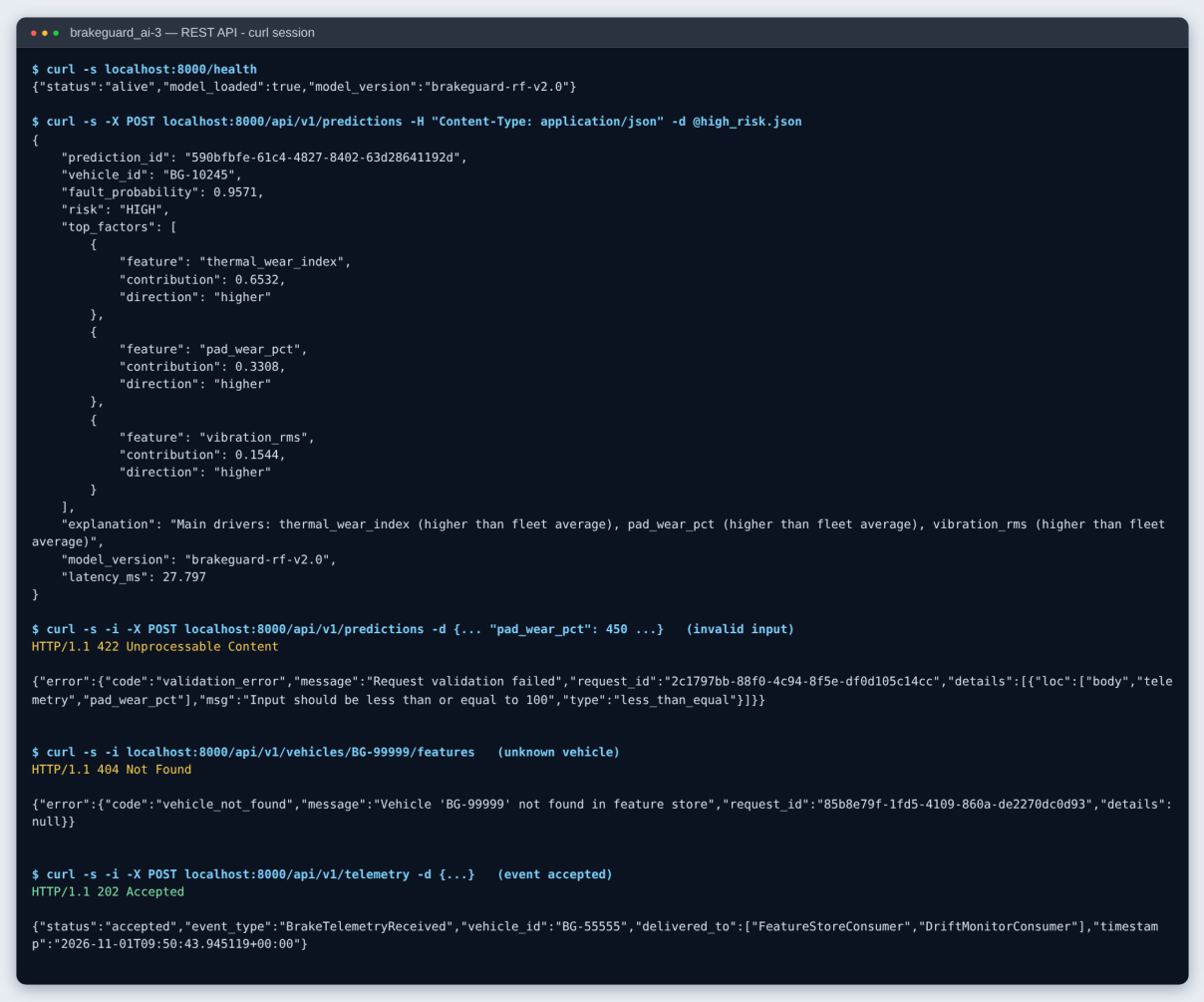


Figure 7 - curl session: 200 prediction with explanation, 422 for pad wear of 450 %, 404 for an unknown vehicle, 202 for a telemetry event.

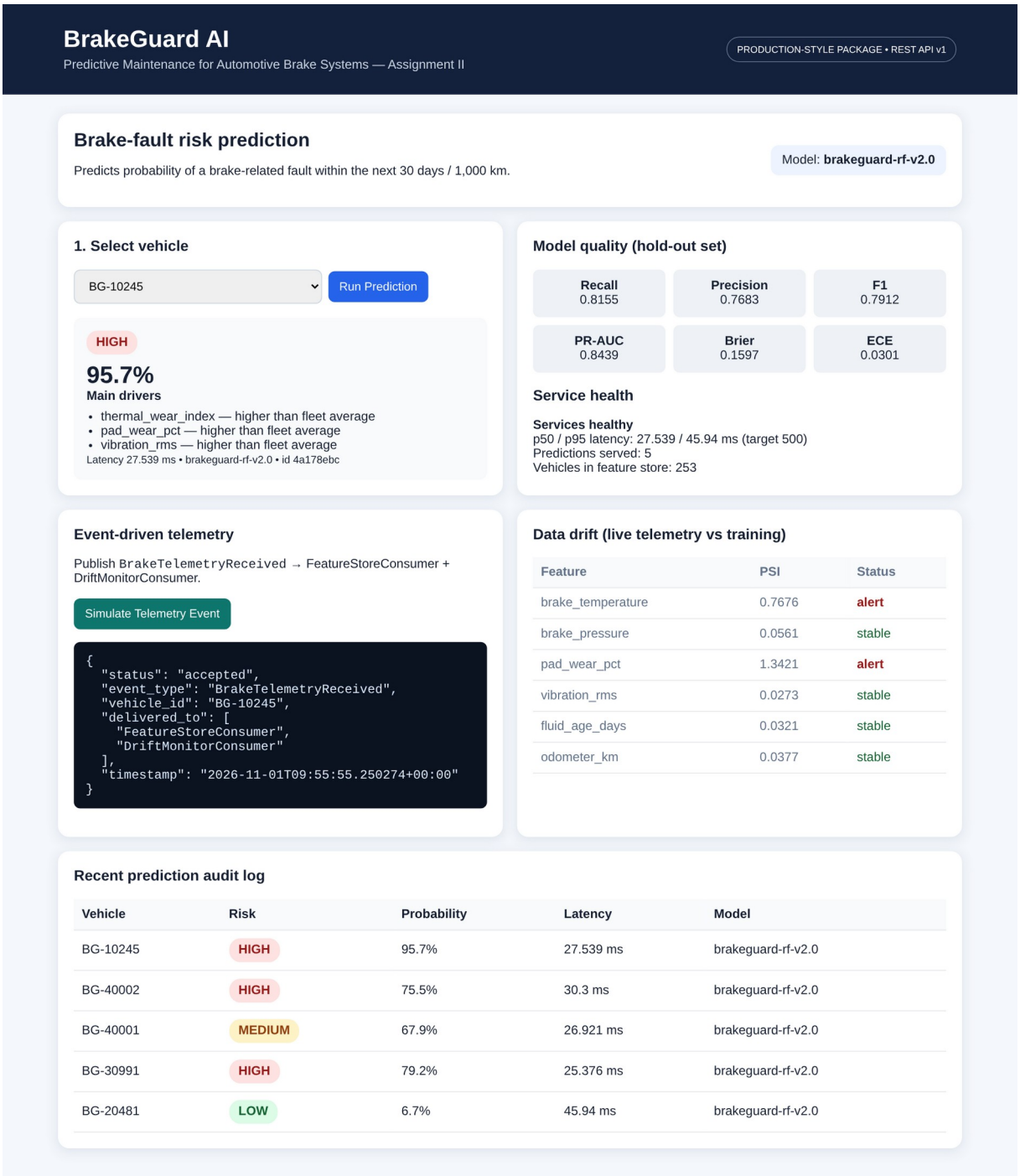


Figure 8 - Dashboard served by the API: prediction with drivers, hold-out model-quality metrics, latency, event-driven telemetry, live drift (a simulated heat-wave fleet drifts on brake temperature and pad wear only) and the audit log.

Objective 2 - Quality assurance

Task 6 - Types of tests (pytest)

The suite has **96 tests in four categories**. All pass in about 12 s, with **97 % line coverage** of the `brakeguard` package. Session-scoped fixtures in `tests/conftest.py` train the production model once. Each API test gets a fresh app instance, so tests never share state.

Type	Location	#	What is verified
Unit tests	tests/unit/	26	Risk-threshold boundaries; FeatureEngineer (derived values, median imputation + WARNING log, missing/non-numeric columns); FeatureStore copies / unknown ids; EventBus isolation of failing subscribers; bounded AuditLog; ECE / metric correctness on hand-computed cases; registry round-trip, missing and corrupt artefacts
Data-validation tests	tests/ data_quality/	19	Schema (missing columns, target), completeness gate, physical-range violations, non-numeric types, invalid labels, duplicates, empty data, CSV ingestion errors, PSI behaviour, DriftDetector on stable vs shifted fleets
ML tests	tests/ml/	24	Training behaviour (8) and inference behaviour (16); see Task 7
Integration tests	tests/integration/	27	HTTP -> services -> model end to end: every endpoint and status code, input-validation and auth rules, event-driven feature-store update, drift monitoring, CSV -> train -> register -> load, bad data blocks registration, retrain fallback, CLI exit codes

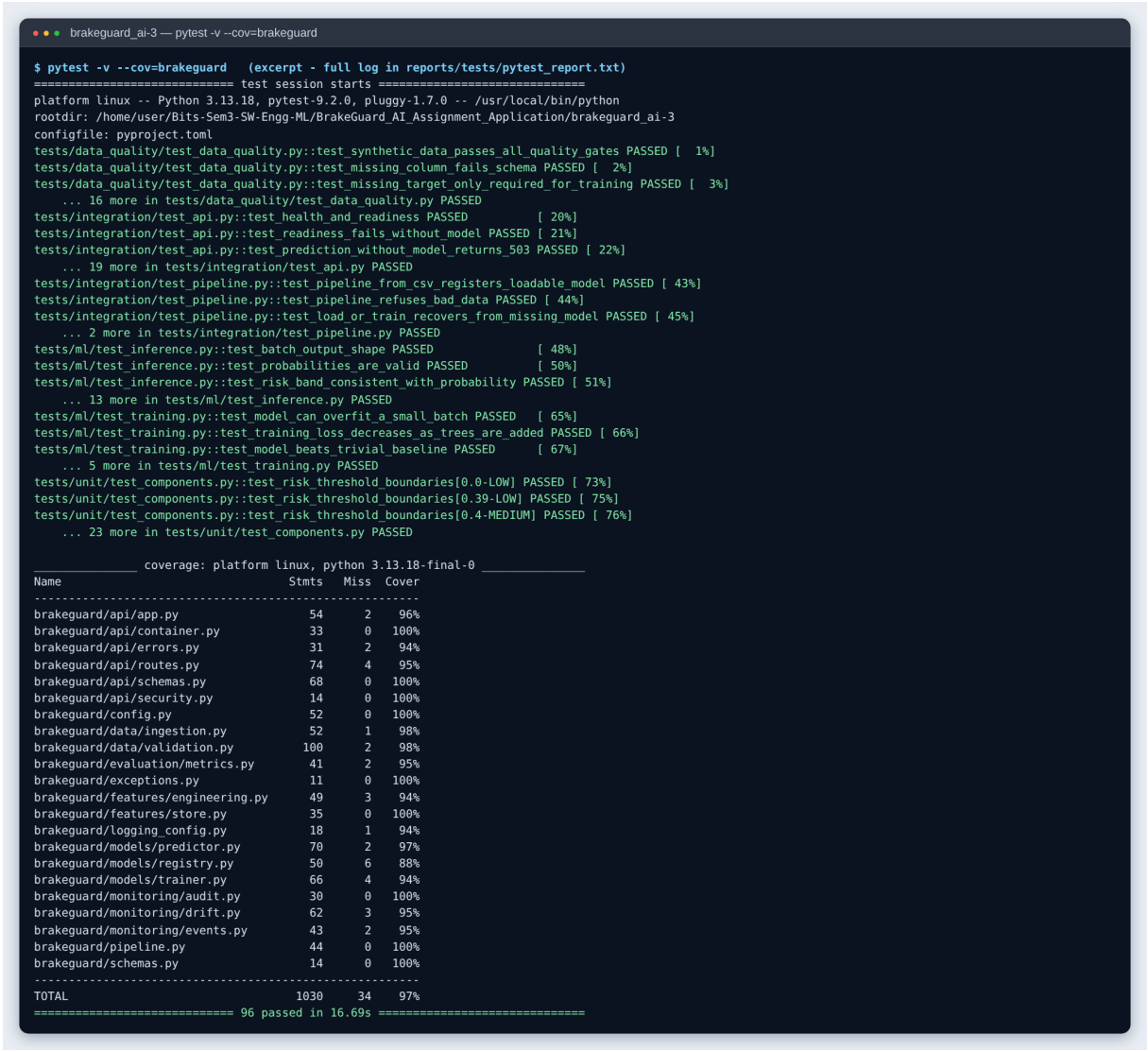


Figure 9 - pytest run (excerpt; full verbose log in reports/tests/pytest_report.txt) with the coverage table.

Task 7 - Tests specific to ML components

7a. Testing model training (tests/ml/test_training.py)

Test	Rationale
`test_model_can_overfit_a_small_batch`	A high-capacity forest must memorise 32 samples (100 % training accuracy). Failure means misaligned labels or dropped features.
`test_training_loss_decreases_as_trees_are_added`	Random forests have no epochs, so the analogue is training log-loss as trees are added (warm_start). It falls from 0.6842 (1 tree) to 0.344 (160 trees) and must never rise by more than 0.01 (Figure 10).
`test_model_beats_trivial_baseline`	PR-AUC must exceed the positive rate (a random classifier's PR-AUC) by 0.2; ROC-AUC > 0.75.
`test_model_meets_quality_targets`	Recall >= 0.75, ECE <= 0.10, Brier <= 0.20.

Test	Rationale
`test_training_is_reproducible`	Same data and seed give identical metrics.
`test_no_severe_overfitting`	Train-test ROC-AUC gap < 0.15.
`test_single_class_target_is_rejected`	Degenerate data raises `ModelTrainingError`.
`test_quality_gate_rejects_model_trained_on_noise`	With shuffled labels the pipeline refuses to register the model and logs an ERROR.

tests/ml/test_training.py

```
def test_model_can_overfit_a_small_batch():
    """A model with enough capacity must memorise 32 samples.

    If it cannot, the pipeline is broken (e.g. labels misaligned with
    features or features silently dropped) - the classic "overfit one
    batch" sanity check.
    """
    batch = SyntheticTelemetrySource(n_samples=32, seed=3).load()
    trainer = ModelTrainer()
    X, y = trainer.split_xy(batch)
    pipeline = trainer.build_pipeline(n_estimators=50, max_depth=None, min_samples_leaf=1,
class_weight=None)
    pipeline.fit(X, y)
    assert (pipeline.predict(X) == y).mean() == 1.0


def test_training_loss_decreases_as_trees_are_added():
    frame = SyntheticTelemetrySource(n_samples=600, seed=5).load()
    trainer = ModelTrainer()
    X, y = trainer.split_xy(frame)
    losses = trainer.loss_curve(X, y, steps=(1, 5, 10, 25, 50, 100))
    assert losses[-1] < losses[0] * 0.5  # big drop from a single tree to the ensemble
    # once converged the curve may plateau, but it must never climb noticeably
    assert all(later <= earlier + 0.01 for earlier, later in zip(losses, losses[1:]))
```

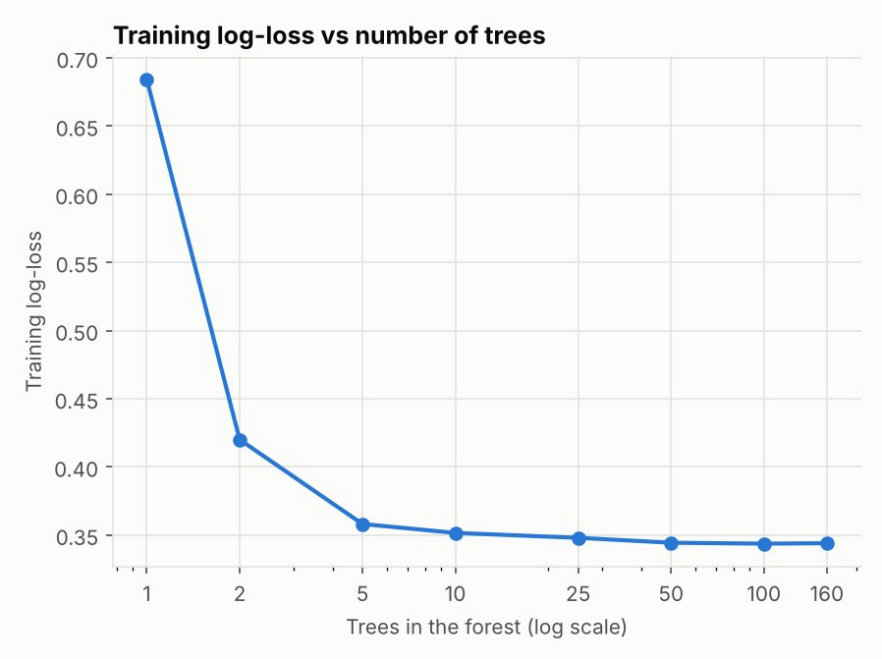


Figure 10 - Training log-loss as trees are added (basis of the 'loss decreases' test).

7b. Testing model inference (tests/ml/test_inference.py)

These tests follow the CheckList approach: shape and range checks, minimum-functionality cases, invariance tests (changes that must not affect the output) and directional-expectation tests (changes that must move the output in a known direction).

Category	Tests
Output shape / range	Batch output shape = (n,); all probabilities in [0, 1] and not NaN; risk band consistent with thresholds; explanation has top-k known features; deterministic; batch == single predictions
Minimum functionality	Obviously failing brakes (230 degC, 92 % wear, 5.5 g vibration) -> HIGH; new brakes -> LOW
Invariance	Key order of the input; extra metadata (vehicle_id, driver name); 0.1 % sensor noise changes probabilities by < 0.01 on average
Directional	Higher pad wear, vibration, temperature or fluid age raises mean fleet probability (> +0.02); line pressure 25 bar or 105 bar both score above nominal 65 bar

tests/ml/test_inference.py - directional expectation tests

```
# -- directional expectations -----------------------------------------------------
@pytest.mark.parametrize(
    ("feature", "low", "high"),
    [
        ("pad_wear_pct", 15.0, 90.0),
        ("vibration_rms", 0.5, 5.0),
        ("brake_temperature", 100.0, 240.0),
        ("fluid_age_days", 30.0, 800.0),
    ],
)
def test_worse_condition_increases_risk(predictor, fleet, feature, low, high):
```

```
    """Averaged over the fleet, a worse value must not lower the fault probability."""
    better = predictor.predict_proba([{**r, feature: low} for r in fleet])
    worse = predictor.predict_proba([{**r, feature: high} for r in fleet])
    assert worse.mean() > better.mean() + 0.02


def test_pressure_deviation_in_either_direction_increases_risk(predictor, fleet):
    nominal = predictor.predict_proba([{**r, "brake_pressure": 65.0} for r in fleet]).mean()
    too_low = predictor.predict_proba([{**r, "brake_pressure": 25.0} for r in fleet]).mean()
    too_high = predictor.predict_proba([{**r, "brake_pressure": 105.0} for r in fleet]).mean()
    assert too_low > nominal and too_high > nominal
```

Task 8 - Model-quality and data-quality metrics

All values below were measured by `python -m scripts.quality_report` (output in `reports/metrics/quality_report.json`) on a 600-row stratified hold-out set of the 3,000-row synthetic fleet data. The same model metrics are stored in the model bundle and served at `/api/v1/model` and on the dashboard.

8a. Model quality

Metric	Why it matters for BrakeGuard	Hold-out	Train	Target / gate
Recall	Share of real brake faults caught; a missed fault is the costly error	0.816	0.861	>= 0.75 (registration gate)
Precision	Share of alerts that are real; controls workshop false alarms	0.768	0.885	monitored
F1	Balance of recall and precision	0.791	0.873	monitored
PR-AUC	Threshold-free ranking quality on the positive class	0.844	0.957	>= 0.70 (registration gate)
ROC-AUC	Threshold-free discrimination	0.847	0.950	> 0.75 (test)
Brier score	Calibration + sharpness of probabilities (lower is better)	0.160	0.101	<= 0.20 (test)
ECE (10 bins)	Does a 70 % shown to a fleet manager mean 70 %?	0.030	0.095	<= 0.10 (test)

Recall is 0.816 at threshold 0.5. Across other synthetic seeds hold-out recall varied between about 0.70 and 0.82, which is why recall is a gate rather than a single trusted number, and why it is re-checked online (Section 9). Calibration is good (ECE 0.030), so the probability shown on the dashboard can be read at face value.

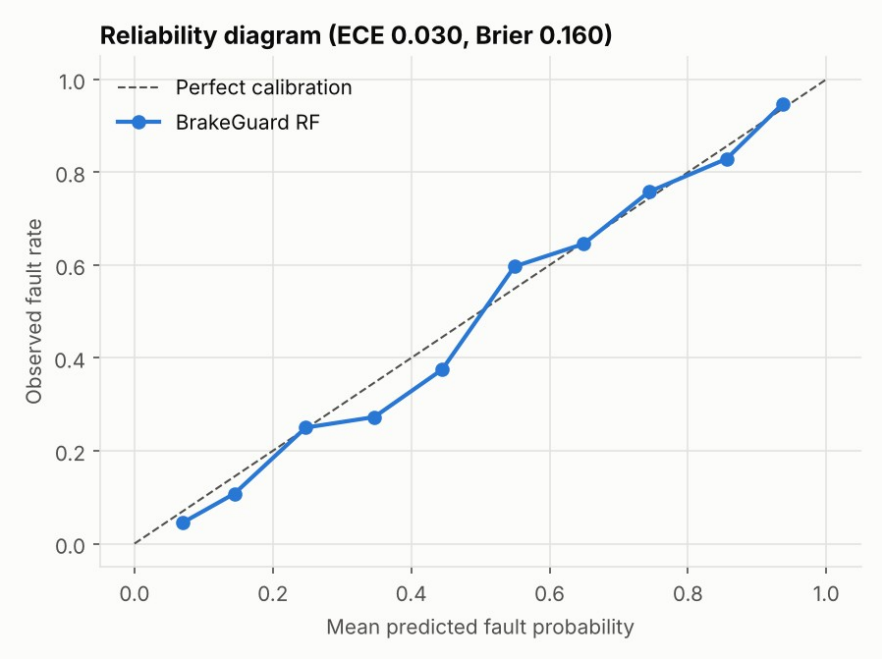


Figure 11 - Reliability diagram: predicted probabilities track observed fault rates.

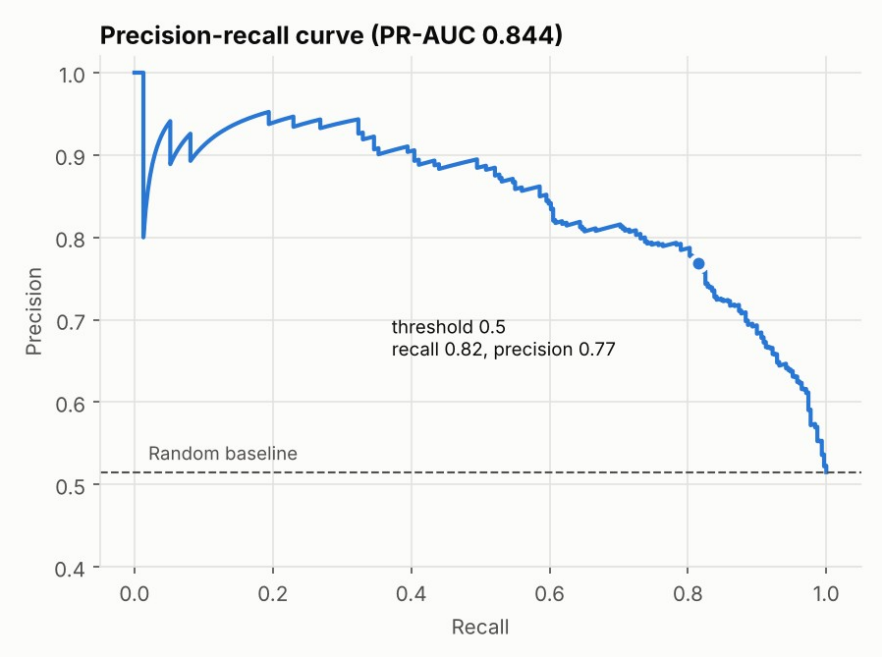


Figure 12 - Precision-recall curve with the operating point at threshold 0.5.

8b. Data quality

`DataValidator` measures data-quality metrics for every batch, and `DriftDetector` compares live telemetry with the training reference. They were run on a clean batch, a deliberately corrupted batch (sensor dropouts, a fraction-vs-percent unit bug and duplicated rows), and a stable and a shifted fleet:

Metric	Definition	Clean batch	Corrupted batch	Gate
Schema validity	All required columns present and numeric	true	true	must be true

Metric	Definition	Clean batch	Corrupted batch	Gate
Missing-value ratio	Share of null cells per column	0.00 %	7.77 % (brake_temperature)	<= 5 % per column
Out-of-range ratio	Values outside the physical range in TELEMETRY_SCHEMA	0.00 %	5.05 % (pad_wear_pct)	<= 1 % per column
Duplicate-row ratio	Exact duplicate records	0.00 %	2.91 %	warning
Gate result		PASSED	REJECTED	

Drift metric	Stable fleet (max over features)	Shifted fleet	Thresholds
PSI (10 reference-quantile bins)	0.021	brake_temperature 1.29, pad_wear_pct 1.45; others <= 0.018	warning 0.10, alert 0.25
Kolmogorov-Smirnov p-value	min 0.139	brake_temperature < 1e-6	< 0.01 confirms shift
Drifted features	none	brake_temperature, pad_wear_pct	

While building the live drift monitor, PSI over 10 bins on a 30-40-sample window flagged every feature as drifting, because the estimate is dominated by noise. The minimum window was therefore made a setting (`BRAKEGUARD_DRIFT_MIN_SAMPLES`, default 200), and an integration test checks that an in-distribution fleet of 250 vehicles raises no drift alert.

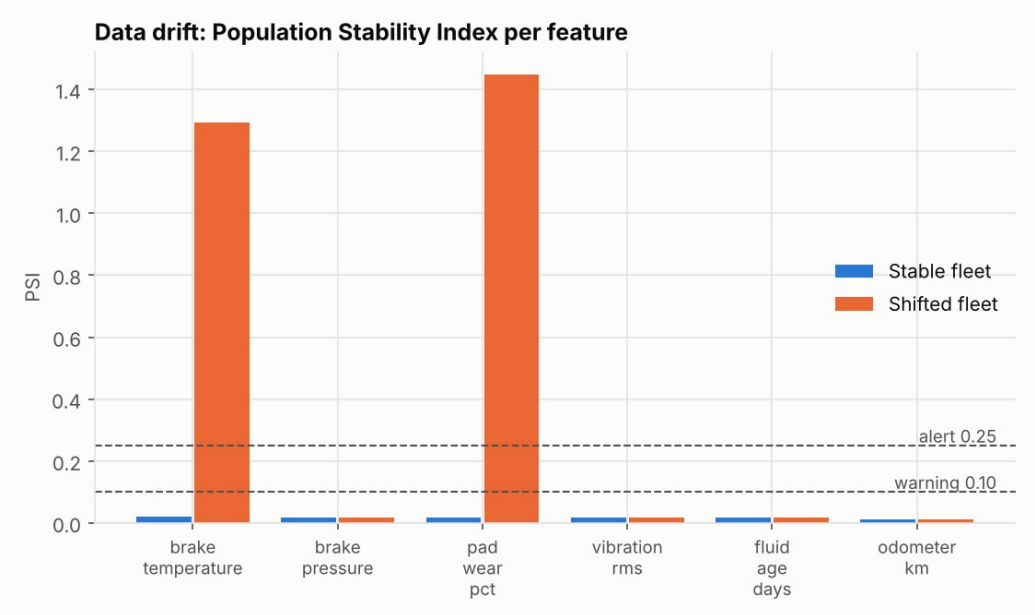


Figure 13 - PSI per feature: the shifted (heat-wave) fleet drifts on temperature and pad wear only.

Task 9 - Testing in production and security

Testing / experimentation in production: shadow -> canary -> A/B

Offline metrics come from synthetic data, and brake faults are only confirmed at the workshop weeks later. A new model version (the challenger) is therefore rolled out against the current champion in three stages:

Stage	Traffic	Exit criteria
1. Shadow (2-4 weeks)	100 % of requests mirrored to the challenger; only the champion's answer is returned and the challenger's is logged	No errors; p95 latency < 500 ms; >= 90 % agreement on HIGH/LOW; challenger recall >= champion on vehicles whose outcome is already known
2. Canary (1-2 weeks)	5 % -> 25 % -> 50 % of vehicles, sticky by vehicle_id hash	Error rate and latency not worse; HIGH-alert rate within +/-20 % of champion; automatic rollback on breach
3. A/B, then full rollout	50/50, then 100 %	After the 30-day label horizon: recall not significantly lower (one-sided test, alpha 0.05), false-alarm rate lower or equal; old version kept in the registry for one-click rollback

Shadow mode comes first because a missed brake fault is a safety incident, so a challenger must not influence a real maintenance decision before its live behaviour is known. The code base already has the needed pieces: `ModelRegistry` versions artefacts, `AuditLog` records `prediction_id` and `model_version` for joining with workshop outcomes, and `/api/v1/monitoring` reports the latency and drift guard-rail metrics that pause or roll back a rollout.

Security consideration: input validation and model access control

Threat. Crafted or faulty telemetry could suppress a real HIGH alert (an evasion attack), poison the feature store and drift statistics, or be used to query the model at high volume to reverse-engineer it. The Assignment I API and the research notebook accept any value; the notebook scores 450 % pad wear as a valid reading.

- **Schema-driven input validation:** physical ranges per field, `extra="forbid"`, a `vehicle_id` pattern and a batch cap of 500. Violations return 422 and are logged as WARNING. Covered by 6 integration tests.
- **Model access control:** when `BRAKEGUARD_API_KEY` is set, `/api/v1/*` requires an `X-API-Key` header, compared in constant time with `hmac.compare_digest`. The key comes only from the environment. Probes stay public for orchestration.
- **Safe failure:** generic 500 responses without stack traces; details are logged under the `request_id`.
- **Bounded state** (audit, event and drift windows) against memory exhaustion, and drift alerts as a detection control for coordinated data injection.
- **Hardened deployment:** multi-stage Docker build with lint and tests as a gate, non-root user, read-only filesystem.

Next steps for production: mTLS from vehicle gateways, per-client rate limiting at the gateway, and signed model artefacts. Unpickling an untrusted joblib file executes code, so only a file whose hash matches the registry entry should be loaded. Full write-up: `docs/production_testing_and_security.md`.

Appendix A - How to run

```
cd BrakeGuard_AI_Assignment_Application/brakeguard_ai-3
python -m venv .venv && source .venv/bin/activate        # Windows: .venv\Scripts\activate
pip install -r requirements-dev.txt
python -m scripts.train --metrics-out reports/metrics/training_metrics.json
uvicorn app:app --reload                                  # dashboard http://127.0.0.1:8000,
Swagger /docs
pytest -v --cov=brakeguard                                # 96 tests
black --check brakeguard tests scripts app.py && isort --check-only brakeguard tests scripts
app.py
flake8 brakeguard tests scripts app.py && pylint brakeguard scripts app.py
python -m scripts.quality_report                          # metrics + figures
docker compose up --build                                 # lint + tests run inside the image
build
```

Appendix B - Deliverables in the repository

Path	Content
brakeguard/	Production package (about 2,000 lines across 29 modules)
tests/	96 pytest tests in unit / data_quality / ml / integration
notebooks/Group14.ipynb	Executed implementation notebook with group details
notebooks/ 01_research_prototype.ipynb	Research-code version used for the Task 2 comparison
docs/	research_vs_production.md, production_testing_and_security.md
reports/lint/	Lint reports: before (Assignment I), before (new package), after
reports/tests/	pytest verbose log + coverage, summary
reports/metrics/	quality_report.json, training_metrics.json, figures
reports/api/	curl transcript, logging evidence
reports/screenshots/	Dashboard, Swagger, lint, tests, API, logging screenshots